

Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

- A. 3.0 amps, 4 ohms
- B. 2.8 amps, 3.2 ohms
- C. 3.75 amps, 3.2 ohms
- D. 4 amps, 3 ohms
- E. 3.33 amps, 3.6 ohms
- F. 5 amps, 2.4 ohms
- G. 4.5 amps, 2.67 ohms
- H. 2.2 amps, 5.45 ohms
- I. 2.5 amps, 4.8 ohms
- J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>

1 . Identify the given information :

- Voltage (V) = 12 volts

- Power (P) = 40 watts

- We need to find current (I) and resistance (R)

2 . Use the formula : Power (P) = Voltage (V) x Current (I) / Resistance (R)

- Rearrange the formula to find Current (I): I = P x R / V

- Plug in the given values : I = 40 watts x R / 12 volts

- Simplify the equation : I = 40 R / 12

- To find the resistance (R), we can use the formula : Resistance (R) = Voltage (V) x Current (I) / Power (P)

- Rearrange the formula to find R : R = V x I / P

- Plug in the given values : R = 12 volts x I / 40 watts

- Simplify the equation : R = 12 I / 40

- We can now substitute the equation for I from the previous step : R = 12 (40 R / 12) / 40

- Simplify the equation : R = 40 R / 40

- Divide both sides by 40 : R = R

- This means that the resistance (R) cancels out . To find the current (I), we can use the equation : I = 40 R / 12

- Plug in the given options for R and solve for I

- We can test each option to see which one satisfies the equation I = 40 R / 12

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